

# Cross-lingual transfer between Azerbaijani and Turkish: Separating Syntactic Transfer, Lexical Warm-Up, and Tokenizer Quality

A pre-registered, failure-aware 164-run study on XLM-15 and XLM-R

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# One task, two adaptation routes

Which route helps a multilingual encoder classify Azerbaijani sentiment — and how reliably?

## Route A — Turkish intermediate training

- Fine-tune on a related high-resource language first (Turkish sentiment, 3 classes), then on Azerbaijani
- Motivated by Kardeş-NLU: Turkish helps low-resource Turkic NLU
- Control: the same Turkish stage with word order shuffled

## Route B — Tokenizer transplantation

- Install an Azerbaijani donor tokenizer (HPLT bert-base-az) and rebuild its embeddings without any training (OMP)
- Removes the segmentation deficit of the original tokenizer
- Controls: two placebo initialisers (Mean, Random) on the same anchors

## This study

- **Task:** binary Azerbaijani sentiment (LocalDoc), low-resource cells  $n = 500$  and  $n = 2,000$ , five seeds each
- **Encoders:** XLM-15 (no Azerbaijani in pretraining) and XLM-R base (Azerbaijani present)
- **Protocol:** hash-locked, pre-registered contrast family; failure-aware reporting; test evaluated once from the restored validation-selected checkpoint

**Narrow, controlled question: does either route beat direct fine-tuning under a fixed 2,000-step budget — and does the answer hold across seeds?**

# Two encoders, two starting points

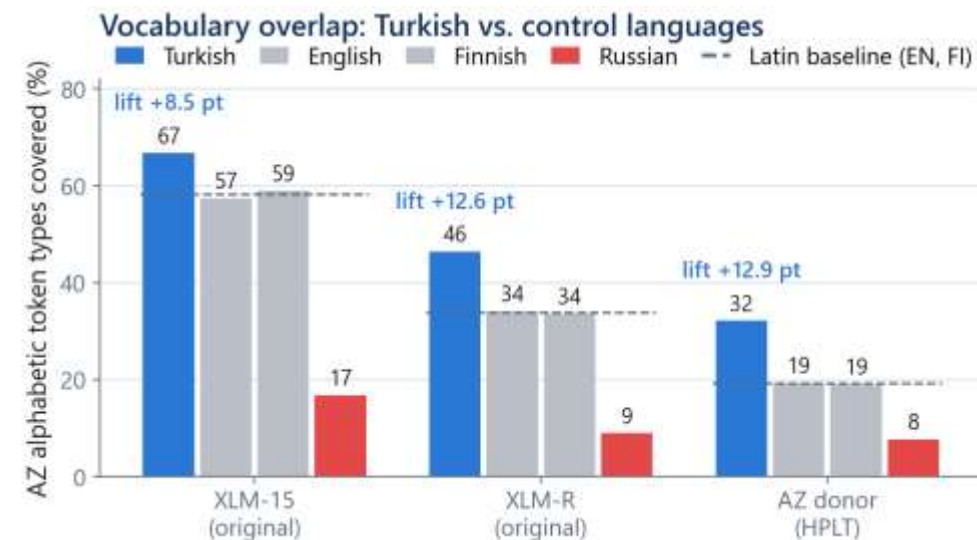
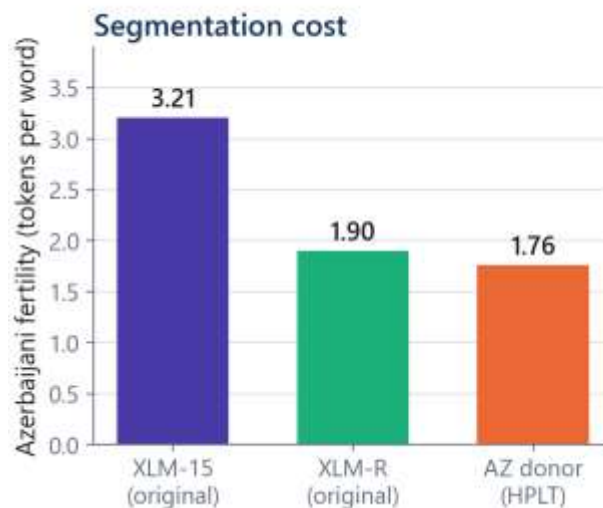
The tokenizer deficit is real for XLM-15; XLM-R already segments Azerbaijani efficiently

## XLM-R base

- Turkish and Azerbaijani in pretraining
- 250k SentencePiece vocabulary
- 1.90 tokens per Azerbaijani word
- Role: strong baseline (contrast)

## XLM-15 (xnli15, 1024-d)

- Turkish yes, Azerbaijani no
- 95k fastBPE vocabulary, no diacritic tokens
- 3.21 tokens per Azerbaijani word
- Role: encoder with a clear deficit (primary)



**Different architectures — cross-encoder comparisons are descriptive only; every inferential contrast stays within one encoder.**

# Experimental design: seven conditions

Every condition ends with the same Azerbaijani fine-tuning stage (head reset, 2,000 steps)

| Condition | Tokenizer / embedding initialisation   | Stage before Azerbaijani          | What it isolates                            |
|-----------|----------------------------------------|-----------------------------------|---------------------------------------------|
| Direct    | Original                               | —                                 | Baseline                                    |
| OMP       | AZ donor / orthogonal matching pursuit | —                                 | The transplant itself                       |
| Mean      | AZ donor / mean of 64 nearest anchors  | —                                 | Placebo: is the OMP solve doing anything?   |
| Random    | AZ donor / random signed coefficients  | —                                 | Placebo: is the anchor span doing anything? |
| TR        | Original                               | Real Turkish sentiment (2 epochs) | Intermediate training                       |
| OMP+TR    | AZ donor / OMP                         | Real Turkish sentiment            | Do the routes combine?                      |
| Shuf. TR  | Original                               | Word-shuffled Turkish             | Order-sensitive part of the Turkish effect  |

**Contrasts of interest: TR vs Direct · TR vs Shuf. TR · OMP vs Mean / Random · OMP+TR vs TR — each a one-factor change.**

# The 164-run grid

Matched primary grid plus a higher-data stress test for the primary encoder

| Encoders         | Conditions                | Training examples<br>n | Seeds | Runs |
|------------------|---------------------------|------------------------|-------|------|
| XLM-15 and XLM-R | all 7                     | 500 · 2,000            | 5     | 140  |
| XLM-15 only      | Direct, OMP, Mean, Random | 10,000 · 20,914        | 3     | 24   |
|                  |                           |                        | Total | 164  |

$$2 \times 7 \times 2 \times 5$$

encoders × conditions ×  
sizes × seeds = 140

$$+ 24$$

XLM-15 extension  
(4 conditions × 2 sizes ×  
3 seeds)

Pre-registered contrast family: all 21 pairwise condition contrasts × 2 encoders × 2 sizes = 84, plus 6 pairwise contrasts × 2 extension sizes = 12 → 96 tests, Holm-corrected as one family.

## Held constant

- 2,000 optimizer steps per Azerbaijani run
- same seeds, same fixed splits
- same validation cadence and selection rule

## What varies

- tokenizer + embedding initialisation
- presence and kind of Turkish stage
- training-set size n

## Extension cells

- XLM-15 only, n = 10,000 and 20,914 (the training split has 20,937 rows)
- seeds {42, 1337, 13}; descriptive stress test

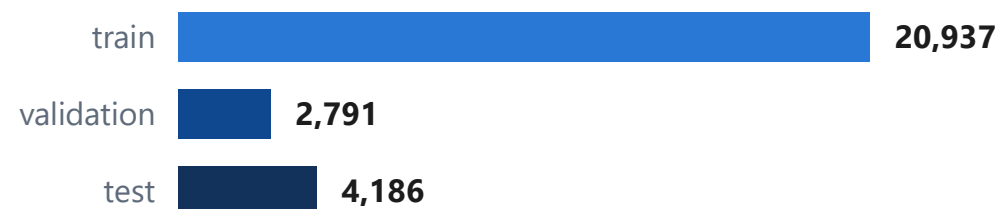
**A fixed step budget holds update opportunity constant across n — so we measure a fixed-budget regime, not converged performance.**

# Datasets, splits and label quality

Fixed, hash-manifested splits; no exact-text leakage between splits

## Azerbaijani — LocalDoc sentiment (Hugging Face)

42,000 three-class texts → neutral removed → 86 duplicates removed → 27,914 binary examples (balanced)



## Turkish — TRSAv1 e-commerce reviews (3 labels)

150,000 → label-aware dedup removes 11,360 → 138,640 → seeded, label-stratified 20,000



### Label-quality audit (two annotators, 100 examples)

- Three-class human–human agreement  $66/99 = 66.67\%$  ( $\kappa = 0.491$ )
- Retained polar rows, unconditioned:  $44/67 = 65.67\%$
- $88.10\%$  ( $\kappa = 0.746$ ) only when all three sources commit to a polarity — a selected, easier subset

### Low-resource cells are deliberate subsamples

- $n = 500$  and  $n = 2,000$  drawn from the fixed 20,937-row training split
- Validation (2,791) drives checkpoint selection; test (4,186) is touched once per run
- Leakage check train/val/test exact-text overlap: 0 / 0 / 0

**Label noise caps attainable macro-F1 — a ceiling in the mid-0.8 range is consistent with 66 % raw human agreement.**

# Transplantation: donor, anchors and three initialisation rules

Donor HPLT bert-base-az (32,770 tokens); coefficients — not vectors — are moved between spaces

For each new donor token  $d$ : pick 256 cosine-nearest donor anchors, then

**OMP**  $\min_{\alpha} \|d - D\alpha\|_2^2, \quad \|\alpha\|_0 \leq 64; \quad b = B\alpha$

**Mean**  $b = \frac{1}{64} \sum_{i \in N_{64}(d)} b_i$

**Random**  $b = \sum_{i \in S_{64}} \alpha_i b_i, \quad \alpha_i = \frac{z_i}{\sum_j |z_j|}, \quad z_i \sim \mathcal{N}(0, 1)$

$D$  = donor vectors of the anchors,  $B$  = the same anchors in the base model;  $b$  is the new base row. Anchors = donor tokens that the base tokenizer encodes as exactly one non-special token (functional match).

**8,047**

functional anchors, XLM-15  
(24.6 % of the donor vocabulary)

**9,776**

functional anchors, XLM-R  
(29.8 % of the donor vocabulary)

## Pipeline safeguards

- Shared rows copied; ~70 % of rows are reconstructions ( $k \leq 64$  of 256 candidates)
- New rows rescaled to the anchor median norm (raw OMP inflated XLM-R norms  $1.91\times$ )
- Tied output rows follow the input embedding; classifier head always reset
- Round-trip serialisation bit-identical; forward-logit deviation 0.0 on reload

**Mean and Random are placebos on the same anchors: if  $OMP \approx \text{Mean} \approx \text{Random}$  downstream, the sparse solve itself adds nothing.**

# Training, selection and the escape criterion

Identical Azerbaijani stage for every condition; test is a one-shot confirmation



| Hyper-parameter                       | Value                                                          |
|---------------------------------------|----------------------------------------------------------------|
| <b>Optimizer</b>                      | AdamW, lr $2 \times 10^{-5}$ , warm-up 10 %, weight decay 0.01 |
| <b>Batch / precision / max length</b> | 32 · fp16 AMP · 256 tokens                                     |
| <b>Budget</b>                         | 2,000 optimizer steps at every n                               |
| <b>Validation cadence</b>             | every 200 steps (10 points incl. terminal)                     |
| <b>Transplant</b>                     | functional anchors, k = 64, m = 256, rescale to anchor median  |

## Escape and pre-registration

- Escape = any recorded validation macro-F1 > 0.40 (a constant predictor scores 1/3 on the balanced split)
- Threshold fixed from pilots and hash-locked with the specification on 4 Sep 2026
- Later changes are dated amendments; none used a test number
- Reported, not excluded: collapsed runs and degenerate Turkish stages stay in the grid

**No model, step or hyper-parameter was ever chosen with a test score — test is a confirmation of validation, nothing more.**

# Three estimands, reported separately

When fine-tuning can collapse, one average hides the failure mode

## Escape probability

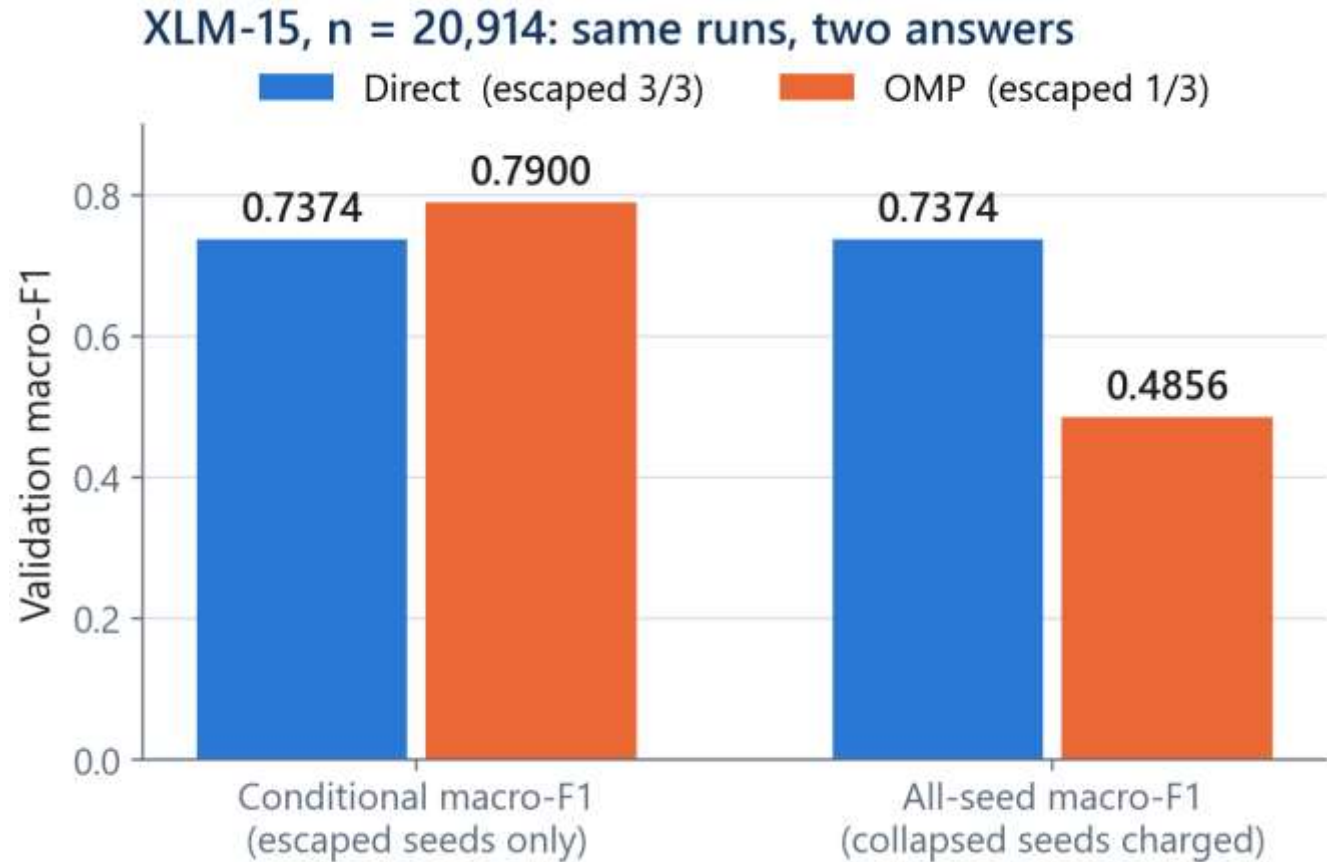
- share of seeds whose validation macro-F1 ever exceeds 0.40 (Wilson 95 % CI)

## Conditional macro-F1

- mean over escaped seeds only — 'how good is it when it works?'

## All-seed macro-F1

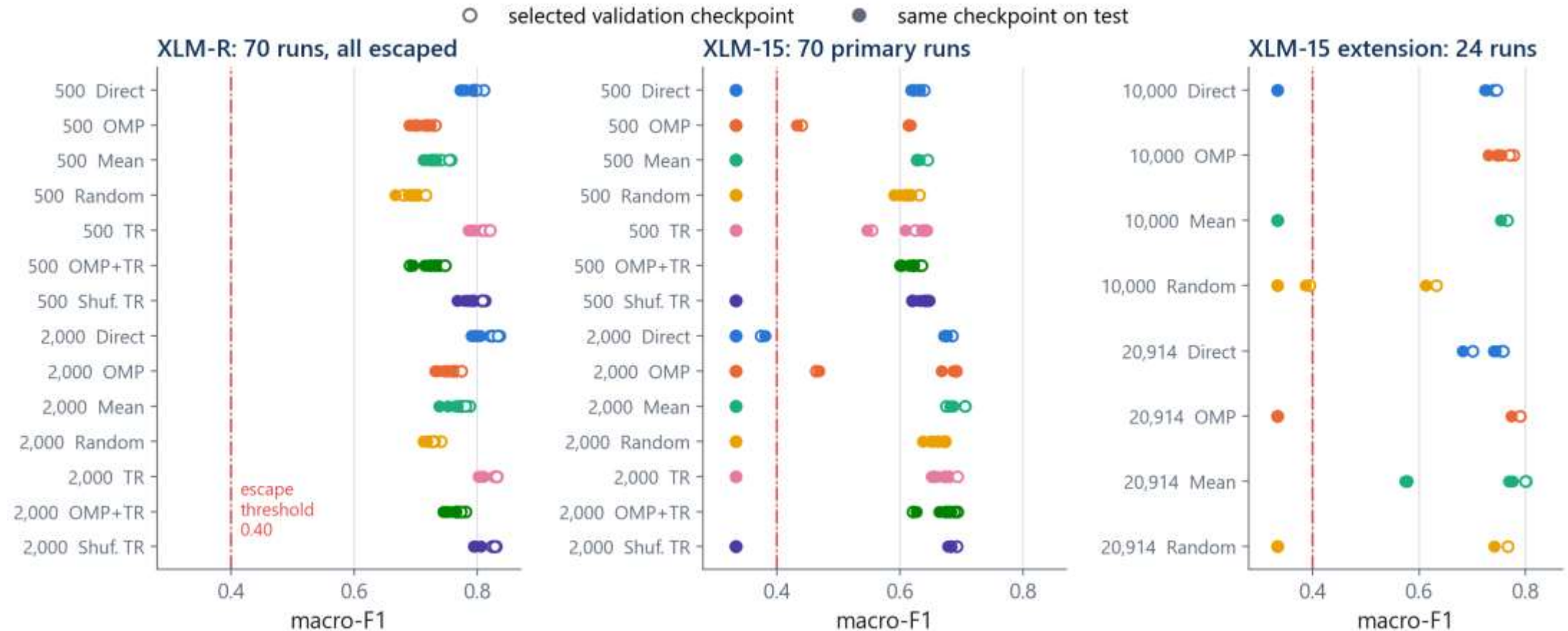
- mean over every declared seed, collapsed seeds charged at their score — 'what do I get on average if I choose this recipe?'



**Same three runs, opposite rankings: conditional F1 favours OMP (0.7900 vs 0.7374), all-seed F1 favours Direct (0.7374 vs 0.4856). Never pool them.**

# Results at a glance: all 164 runs

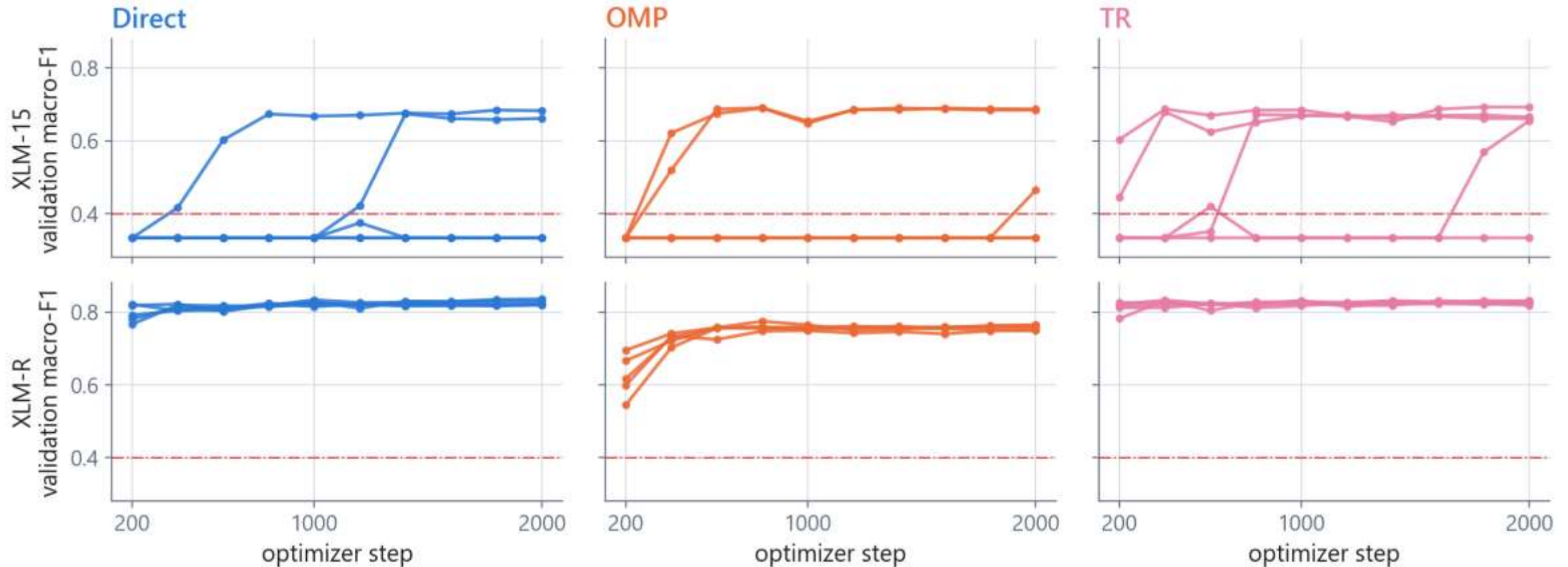
Open marker = selected validation checkpoint; filled = the same checkpoint on test; one row per cell



**XLM-R: every run escapes and the original-tokenizer conditions sit on top. XLM-15: a cluster at  $\approx 0.33$  is collapse, not a weak model.**

# What collapse looks like during training

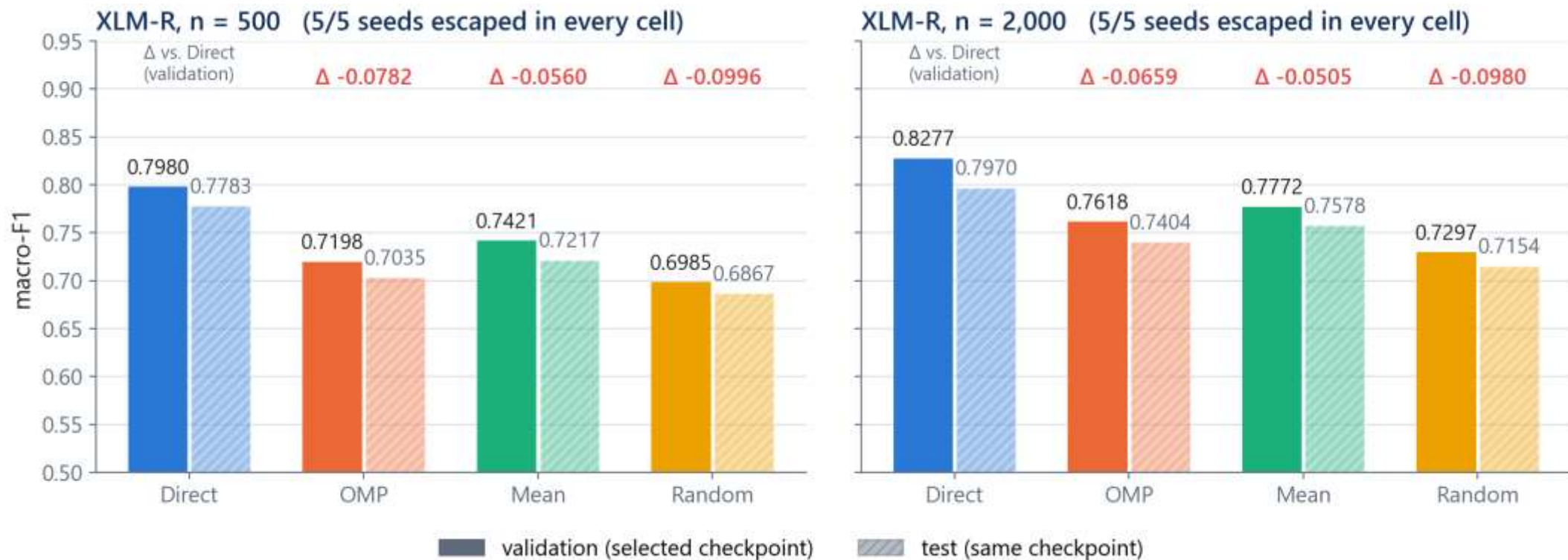
Validation macro-F1 every 200 steps,  $n = 2,000$ , five seeds per panel



**XLM-R seeds separate from 1/3 within 200 steps; several XLM-15 seeds never leave the constant-predictor plateau within the 2,000-step budget.**

# Headline result: the implemented transplant harms XLM-R

Validation and test,  $n = 500$  and  $n = 2,000$ ; direction identical in every seed

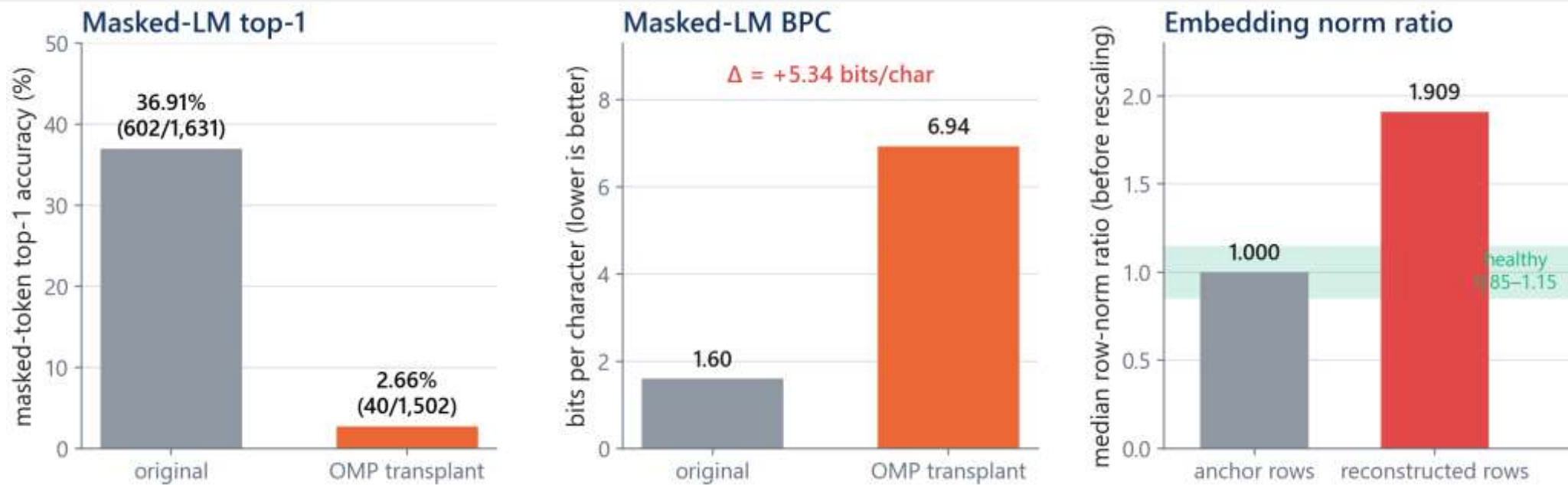


**Per-seed Direct  $\rightarrow$  OMP validation  $\Delta$  at  $n = 2,000$ :** -0.074, -0.060, -0.061, -0.075, -0.060 (5/5 negative); **at  $n = 500$ :** -0.096, -0.071, -0.094, -0.066, -0.064. Test confirms: 0.7970  $\rightarrow$  0.7404 at  $n = 2,000$ .

**Ordering on XLM-R is Direct  $\approx$  TR  $>$  Mean  $>$  OMP  $>$  Random: the crude anchor mean is at least as good as the OMP solve, and both lose to the original tokenizer.**

# Why it fails: three intrinsic probes agree

XLM-R original vs. the installed OMP transplant, 300 training lines, masked-LM diagnostics



## What the probes show

Masked-token top-1 36.91 %  $\rightarrow$  2.66 %; +5.34 bits/char; raw reconstruction inflated norms  $1.91\times$  (rescaled before training). The transplanted head is nearly blind to Azerbaijani.

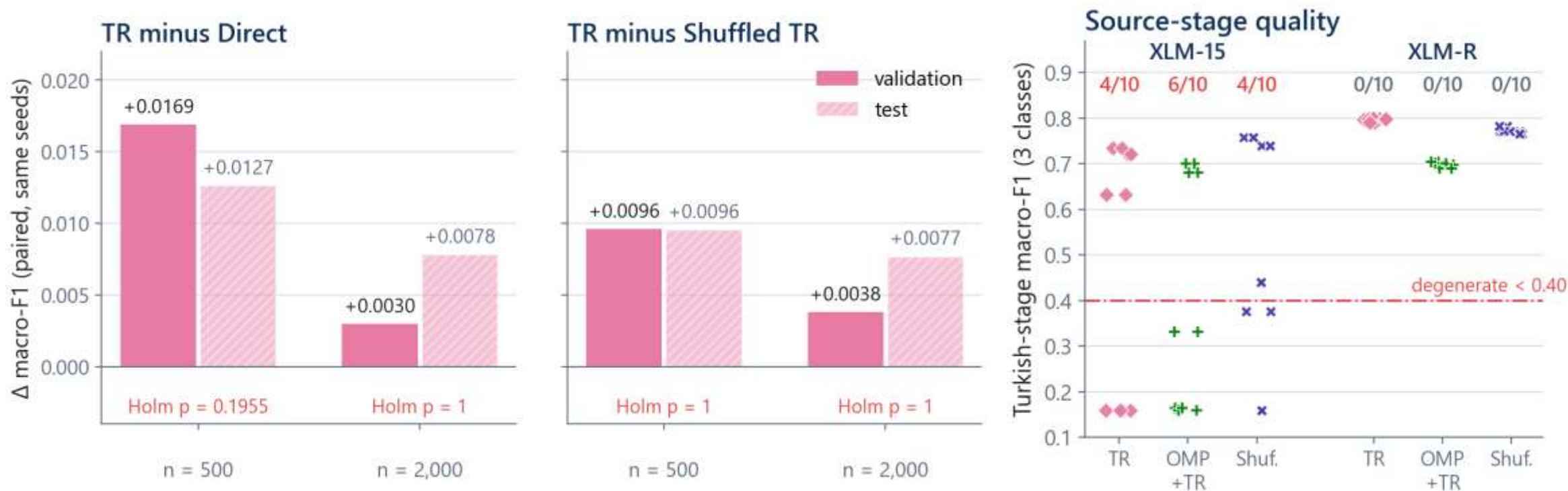
## What they cannot show

Where the defect sits: anchor coverage (30 %), the 64-sparse solve across dimensions, rescaling, changed segmentation or tied output rows all change at once. Implementation-bounded, not a verdict on OMP in general.

**Serialisation is ruled out (bit-identical round trip, zero logit deviation) — the damage is in the reconstructed embeddings, not in the plumbing.**

# Turkish stage: numerically best, statistically unresolved

XLM-R paired contrasts (same seeds) — and why XLM-15 cannot be assessed

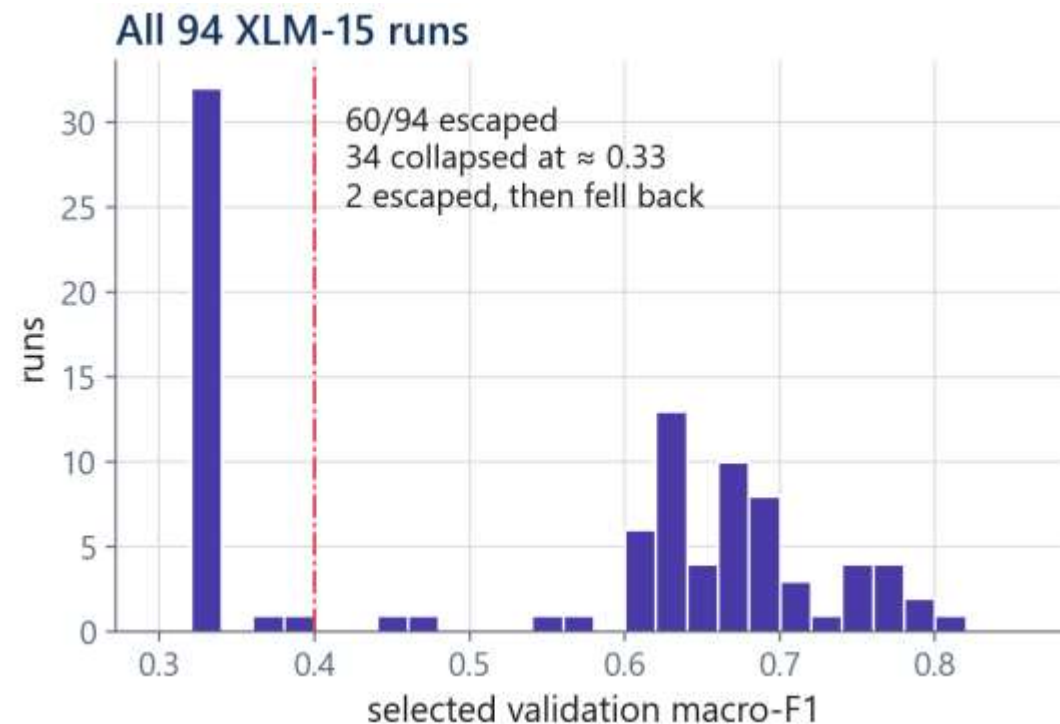
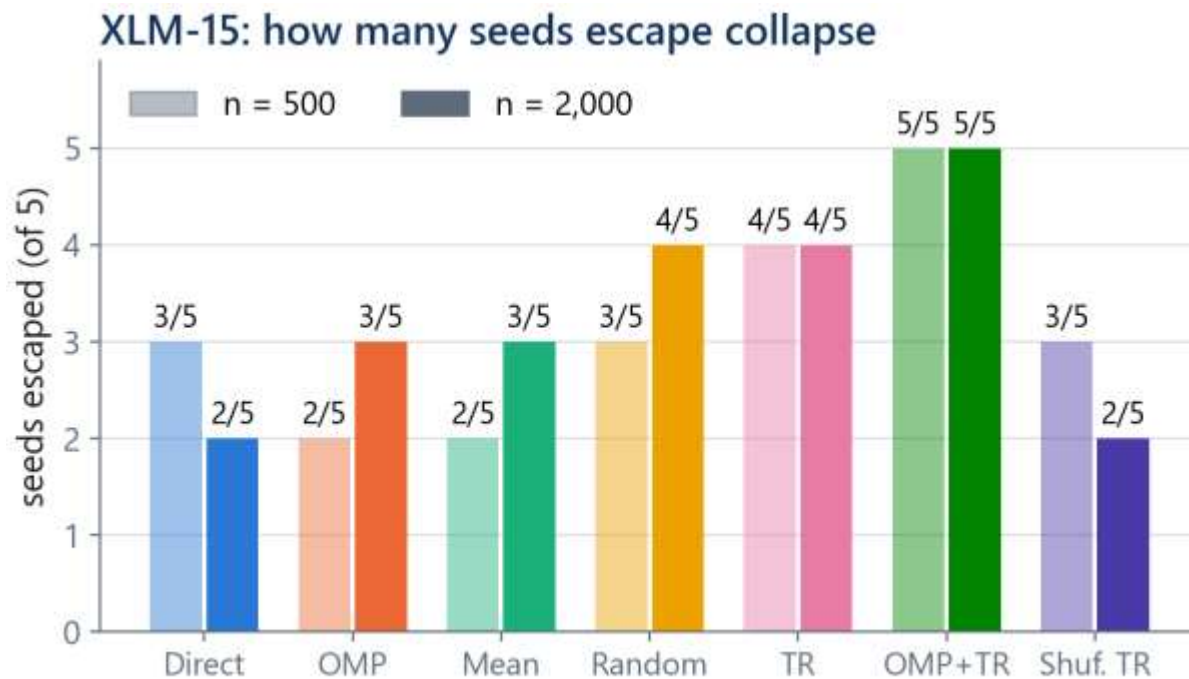


**TR is the best XLM-R condition at both sizes (0.8149 / 0.8307 validation; 0.7910 / 0.8048 test).** The  $n = 500$  gain (+0.0169, Welch  $p = 0.0034$ ) becomes Holm  $p = 0.1955$  over the 86-test family. Real-vs-shuffled Turkish: +0.0096 and +0.0038 — so the order-sensitive part, if any, is smaller than the whole effect. On XLM-15, 14 of 30 Turkish stages are degenerate (1–2 of 3 classes predicted): the treatment was often not delivered.

**Direction agrees with Kardeş-NLU's low-resource claim; magnitude is modest and no Direct-relative Turkish gain survives correction.**

# The escape problem dominates XLM-15

60 of 94 XLM-15 runs escape; the rest predict a single class

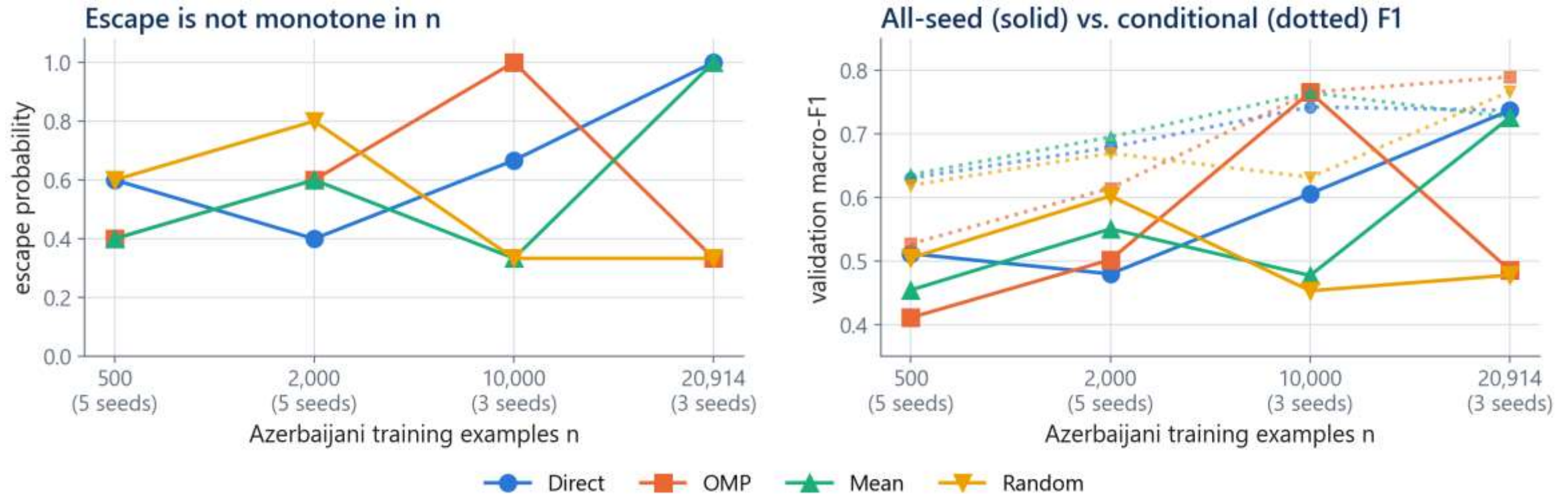


**Different conditions keep different seeds**, so conditional means compare selected survivors (e.g.  $n = 2,000$ : Direct 2/5 vs OMP+TR 5/5). Five seeds cannot separate 2/5 from 5/5 at conventional significance (Fisher). Two runs escaped and later fell back below 0.40; restoring the selected checkpoint keeps their escaped score.

**No XLM-15 contrast survives correction: with collapse this frequent, the honest statement is about reliability, not about which recipe is best.**

# More data does not fix XLM-15 monotonically

Extension cells  $n = 10,000$  and  $20,914$  (three seeds) under the same 2,000-step budget



Escape counts per condition across  $n$ : Direct  $3/5 \rightarrow 2/5 \rightarrow 2/3 \rightarrow 3/3$ ; OMP  $2/5 \rightarrow 3/5 \rightarrow 3/3 \rightarrow 1/3$ ; Mean  $2/5 \rightarrow 3/5 \rightarrow 1/3 \rightarrow 3/3$ ; Random  $3/5 \rightarrow 4/5 \rightarrow 1/3 \rightarrow 1/3$ . With a fixed step budget, larger  $n$  means fewer passes over the data ( $\approx 125$  epochs at  $n = 500$ ,  $\approx 3$  at  $n = 20,914$ ), so this is a property of the regime, not of the data.

**The stress test is descriptive (3 seeds): it shows the collapse pattern is not a small- $n$  artefact and that conditional F1 alone would mislead at every  $n$ .**

# Statistical discipline: one family, one correction, one exact floor

Pre-registered Welch contrasts, Holm over the whole family, exact paired sign-flip as the floor

96

pre-registered contrasts

86

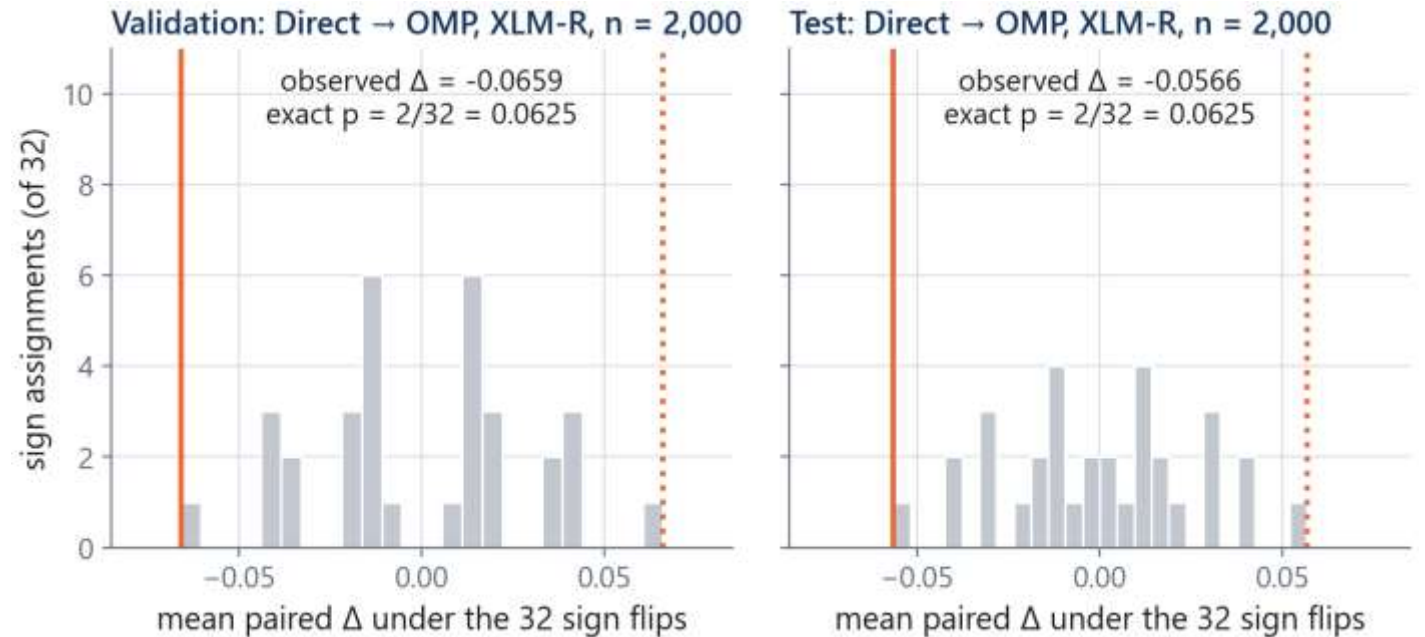
measurable ( $\geq 2$  escaped seeds on both sides)

27

survive Holm — all on XLM-R

27/27

keep sign and significance on test

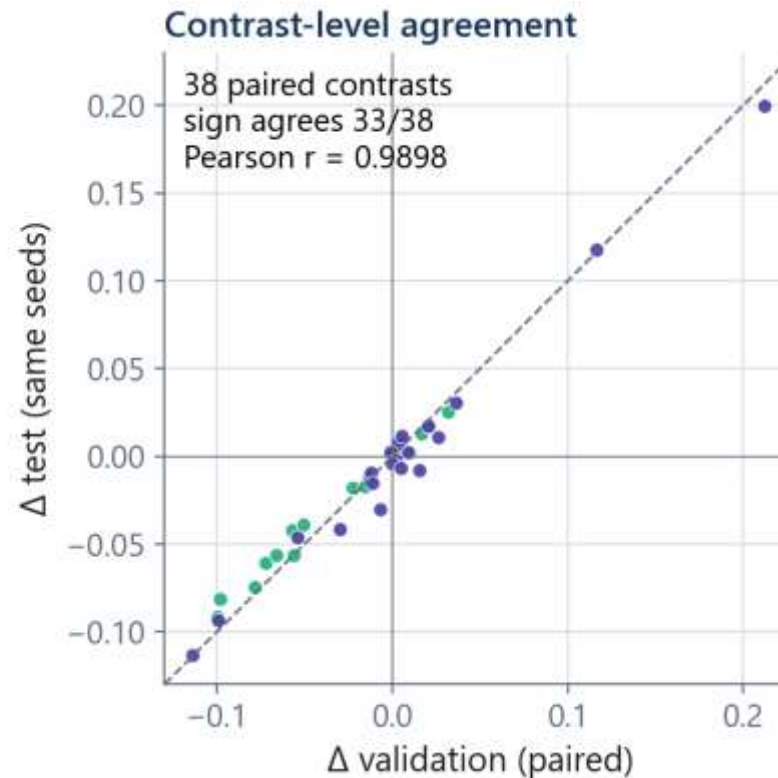
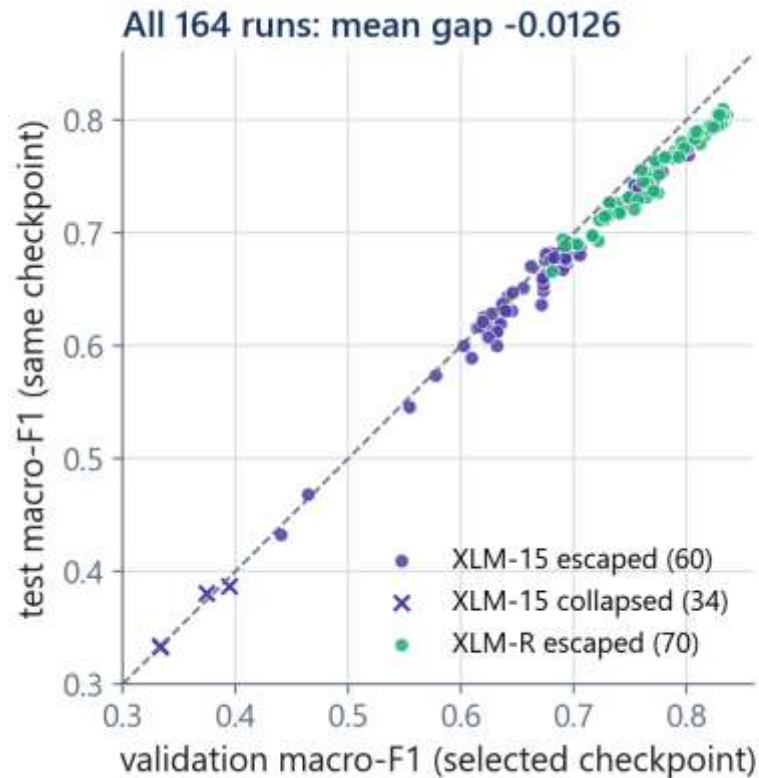


**Five seeds  $\Rightarrow 2^5 = 32$  sign assignments  $\Rightarrow$  smallest two-sided exact p =  $2/32 = 0.0625$ .** Survivors: transplants vs. Direct (all four at  $n = 2,000$ ; OMP, Mean, Random at  $n = 500$ ); every transplant vs. TR and Shuf. TR at both sizes; three Random-placebo contrasts. Not surviving: OMP vs Mean, TR vs Direct.

**27 survivors are model-based statements; the design's exact floor (0.0625) is stated with every one of them.**

# Validation predicts test

Test scores come from the restored validation-selected checkpoint; they were never used for selection



**-0.0126**

mean test – validation gap  
(matched weights, 164 runs)

**$r = 0.9898$**

contrast-level agreement  
(38 paired contrasts)

**0.874**

test / validation effect shrinkage  
(confirmation, not inflation)

Collapsed runs have gap  $\approx 0$  (constant predictor on both splits); escaped runs lose  $\approx 0.016$  on test, consistently across both encoders.

**Test confirms validation everywhere: 27/27 survivors keep their direction; effects shrink slightly, as expected without test-based selection.**

# Compute envelope and reproducibility

Well inside the brief's ~2-day / 10–12 GB budget; every number traces to a file in the repository

## 164

completed runs, schema v4  
(one JSON record each)

## 8.36 h

process-hours summed over runs  
(≈ 9.5 GPU-h wall-clock incl. Turkish  
cache)

## ≈ 3.1 min

mean run time  
(min 2.5, max 6.8)

## 1 × H100

NVL, single stream throughout  
(8.6 GB peak on an RTX 4060 pre-grid  
benchmark)

### Reproducibility chain

- Specification hash-locked 4 Sep 2026 (configs/CONFIG\_HASHES.lock); amendments dated and justified
- Split content manifests (SHA-256) and seeds fixed; donor pinned to revision a126552c
- aggregate.csv → summary.csv → stats.csv regenerated from the run records; paper tables and figures generated by scripts, not typed
- Repository tag v1.0-final: [github.com/shahinsafarli/az-tokenizer-transfer](https://github.com/shahinsafarli/az-tokenizer-transfer)

### Stack

- PyTorch 2.11.0 + CUDA 12.8 build on CUDA 13.2 host; Hugging Face Transformers / Tokenizers
- Bases: FacebookAI/xlm-mlm-tlm-xnli15-1024 and xlm-roberta-base; donor HPLT/hplt\_bert\_base\_az
- Statistics: Welch t, Holm, Wilson CIs, bootstrap CIs, exact paired sign-flip
- Not measured: peak GPU memory during the grid (sampler did not run) — stated, not estimated

**Everything shown today can be regenerated from a clean checkout; where a quantity was not measured, the paper says so.**

# What we can and cannot claim

Scope of the evidence, stated before the conclusions

## Supported by the data

- This OMP implementation, on these encoders, with these anchors and this budget, does not recover the original-tokenizer result on XLM-R
- The three initialisers are ordered Mean > OMP > Random downstream on XLM-R — the anchor span, not the solver, looks limiting
- Turkish intermediate training is numerically best on XLM-R at both sizes
- XLM-15 outcomes are dominated by optimisation failure under a 2,000-step budget
- Validation-based conclusions transfer to test without inflation

## Not established — and why

- That OMP transplantation is harmful in general (single implementation; no continued pretraining after transplant)
- That Turkish training helps (Holm  $p = 0.1955$  at  $n = 500$ ; exact floor 0.0625 with five seeds)
- Anything about XLM-15's Turkish route: 14/30 source stages degenerate, most contrasts rest on 2–3 seeds
- Converged performance: a fixed step budget measures a fixed-budget regime
- Generality beyond one binary sentiment task with  $\approx 66\%$  human agreement

**Narrow negative results on one task do not refute Kardeş-NLU or the tokenizer-adaptation literature — they bound one implementation under one budget.**

# Conclusion

164 completed runs · failure-aware reporting · restored-checkpoint testing

| Question                                                | Observed result                                                                                                | Interpretation                                                   |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| <b>Does the OMP transplant help XLM-R?</b>              | Consistent harm: 0.8277 → 0.7618 (val), 0.7970 → 0.7404 (test) at n = 2,000; 5/5 seeds; intrinsic probes agree | Implementation-bounded negative result; placebo Mean $\geq$ OMP  |
| <b>Does Turkish intermediate training help?</b>         | Numerically best on XLM-R (+0.0169 at n = 500, +0.0030 at n = 2,000)                                           | Direction plausible, advantage over Direct unresolved after Holm |
| <b>What happens on the encoder without Azerbaijani?</b> | XLM-15: 60/94 escapes; rankings depend on escape and n                                                         | Report escape, conditional and all-seed F1 — never one number    |
| <b>How much survives strict inference?</b>              | 27 of 86 contrasts (all XLM-R); 27/27 confirmed on test; exact floor p = 0.0625                                | Model-based survivors with a stated design floor                 |
| <b>Scope</b>                                            | One binary sentiment task, two encoders, fixed 2,000-step budget                                               | No refutation of Kardeş-NLU; a bounded, reproducible measurement |

**The honest contribution is methodological as much as empirical: pre-registration, placebos, and failure-aware estimands changed what we could claim.**

*Every number on these slides is in the report and traces to results/aggregate.csv, summary.csv, stats.csv, the intrinsic-probe JSONs and the received test tables.*

# Thank you - questions welcome

Comparing Turkish intermediate training and tokenizer transplantation for Azerbaijani sentiment classification

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